

**Emergency Department Volume and Capacity
Using Python**

Final Project Summary

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Project Overview

Healthcare systems often struggle with managing Emergency Department (ED) demand due to increasing patient volume and limited resources. This project analyzes patient volume, hospital capacity, and resource utilization to identify inefficiencies and improve decision-making.

Data Source

Dataset Name	Emergency Department Volume & Capacity Analysis
Source	Data.gov
File Type	csv
Main Dataset	Healthcare
Number of Records	12,849
Number of Attributes	18

Business Problem

Hospitals need to address several critical operational questions to ensure efficient healthcare delivery. They must evaluate whether emergency departments are overloaded, assess if there are sufficient beds and treatment stations, and identify which regions experience higher patient demand. Additionally, it is important to determine whether there are shortages in primary care or mental health services and explore ways to optimize resources to reduce patient waiting times. Without proper data analysis and planning, these challenges can lead to serious issues such as overcrowding, delayed treatments, and poor resource allocation, ultimately affecting the quality of patient care.

Objectives

- Analyze patient volume across hospitals
- Evaluate hospital capacity (beds & ED stations)
- Detect overcrowding and high-demand facilities
- Compare urban vs rural healthcare demand
- Identify shortages in primary care & mental health
- Study trends over time (2021–2023)
- Provide data-driven insights for optimization

Dataset Attributes

Original Attribute	Renamed Column	Description
oshpd_id	hospital_id	Unique ID for each hospital facility
FacilityName2	hospital_name	Name of the hospital
CountyName	county	County where the hospital is located
system	hospital_system	Hospital network/system affiliation
LICENSED_BED_SIZE	bed_size	Category of licensed bed capacity
HospitalOwnership	ownership	Type of hospital ownership
UrbanRuralDesi	area_type	Indicates whether hospital is in urban or rural area
TEACHINGDesignation	teaching_status	Teaching or non-teaching hospital
Category	condition	Health condition category
Tot_ED_NmbVsts	total_visits	Total number of emergency visits
EDStations	ed_stations	Number of emergency department stations
EDDXCount	condition_visits	Number of visits for specific condition
LATITUDE	latitude	Latitude of hospital location
LONGITUDE	longitude	Longitude of hospital location
PrimaryCareShortageArea	primary_care_area	Indicates if area has primary care shortage
MentalHealthShortageArea	mental_health_area	Indicates if area has mental health shortage
Visits_Per_Station	visits_per_station	Average visits handled per station
Year	year	Year of the data

Tools & Technologies

-  Python
-  Google Colab
-  Pandas
-  NumPy
-  Matplotlib
-  Seaborn

Data Preprocessing & Feature Engineering

- 🛠 Examined dataset using first and last rows
- 🛠 Performed statistical summary for numerical understanding
- 🛠 Renamed columns for better readability
- 🛠 Analyzed categorical columns using unique values
- 🛠 Identified and removed duplicate records
- 🛠 Checked and handled missing values
- 🛠 Evaluated emergency department load
- 🛠 Analyzed hospital capacity (beds & treatment stations)
- 🛠 Identified regions with high patient demand
- 🛠 Checked for healthcare facility shortages
- 🛠 Suggested resource optimization based on insights

Statistical Analysis

I performed statistical analysis on the **total_visits** column using measures like mean, median, mode, variance, and standard deviation to understand the distribution and variability of patient visits.

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Skewness
- Kurtosis

Exploratory Data Analysis (EDA)

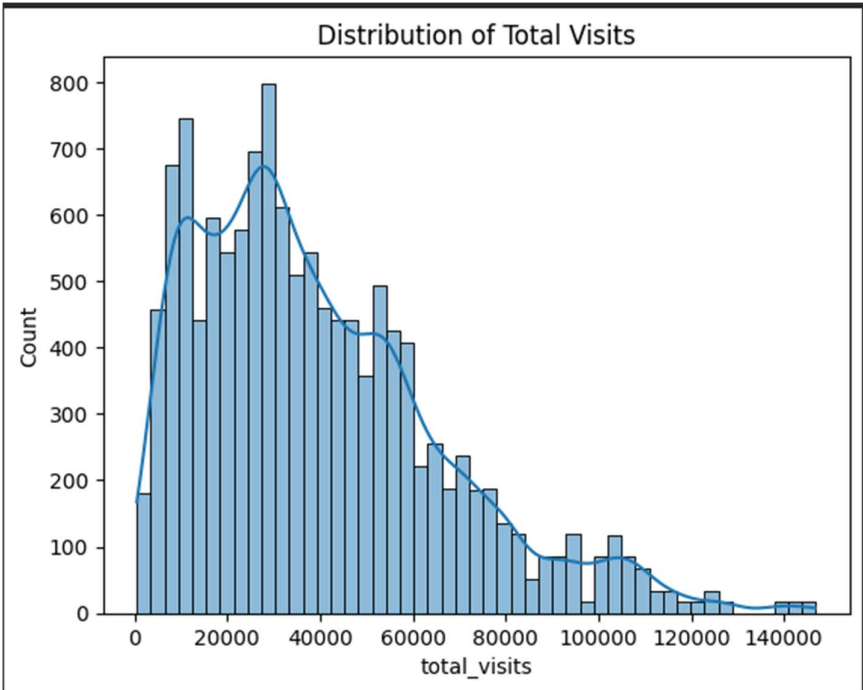
Exploratory Data Analysis (EDA) was performed to understand the dataset, identify patterns, and detect anomalies in emergency department visits.

- Checked data types and structure
- Handled missing values
- Analyzed distributions using histograms & boxplots
- Identified outliers in **total_visits**
- Explored relationships using scatterplots & correlation

Data Visualizations

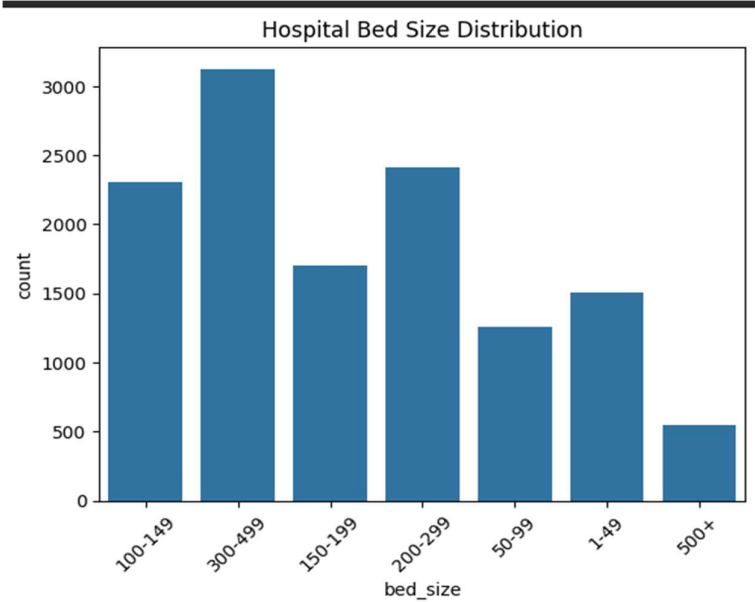
Univariate Analysis

1. Distribution of Total Visits



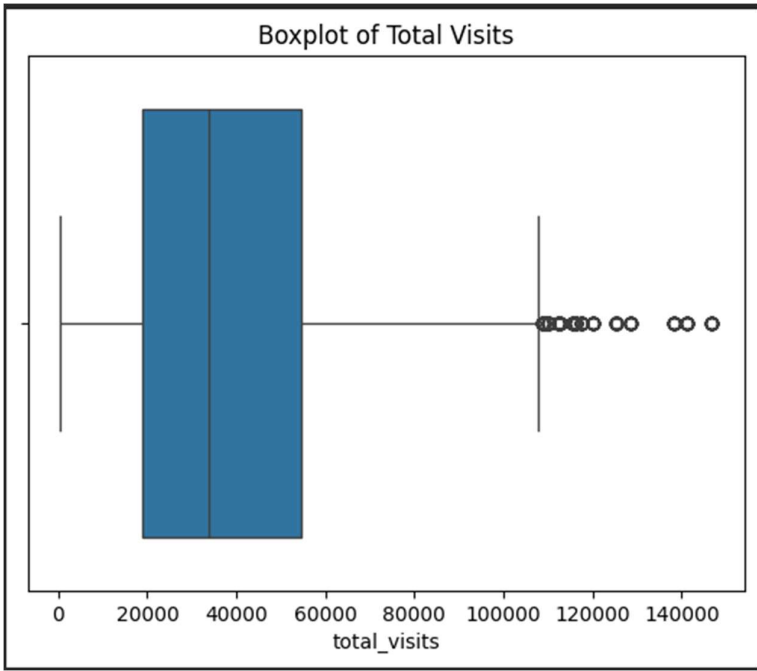
Interpretation: Most hospitals have moderate visits

2. Hospital Bed Size Distribution



Interpretation: Most hospitals are medium-sized (300–499 beds), very large hospitals are fewer

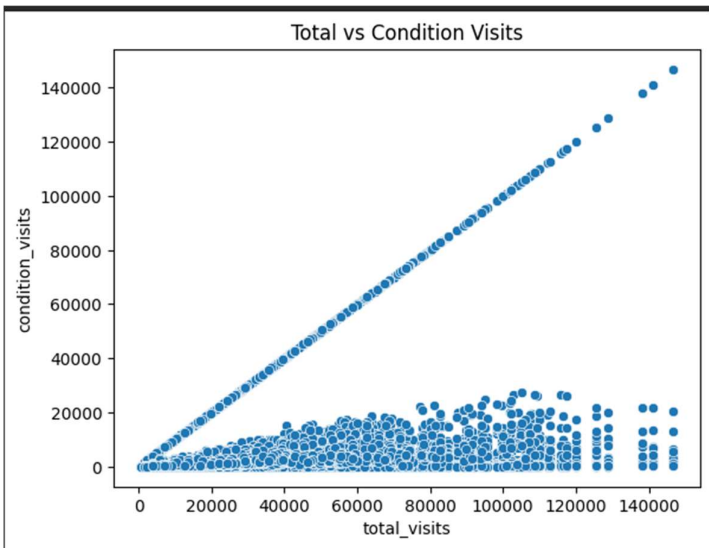
3. Outlier Detection in Patient Visits



Interpretation: Most hospitals have moderate visits, but a few hospitals have extremely high visits (outliers)

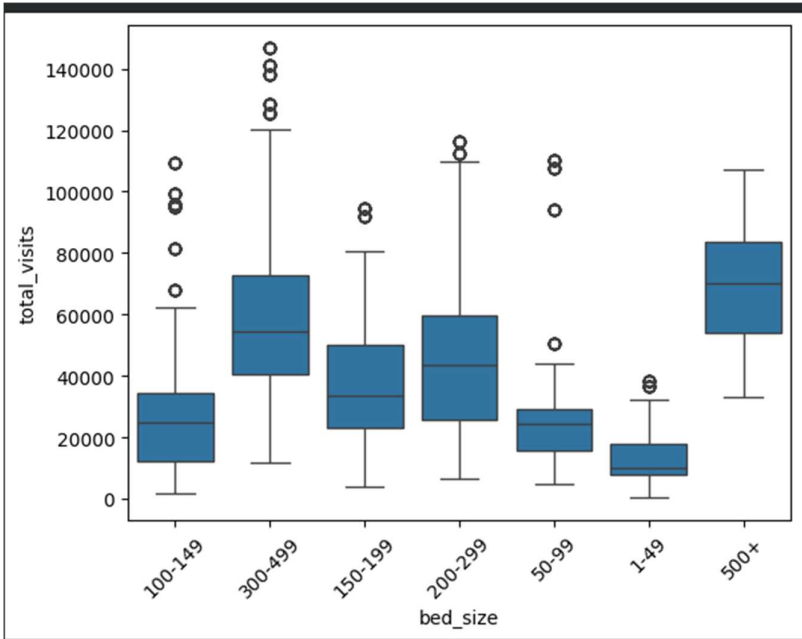
Bivariate Analysis

1.Total Visits vs Condition Visits



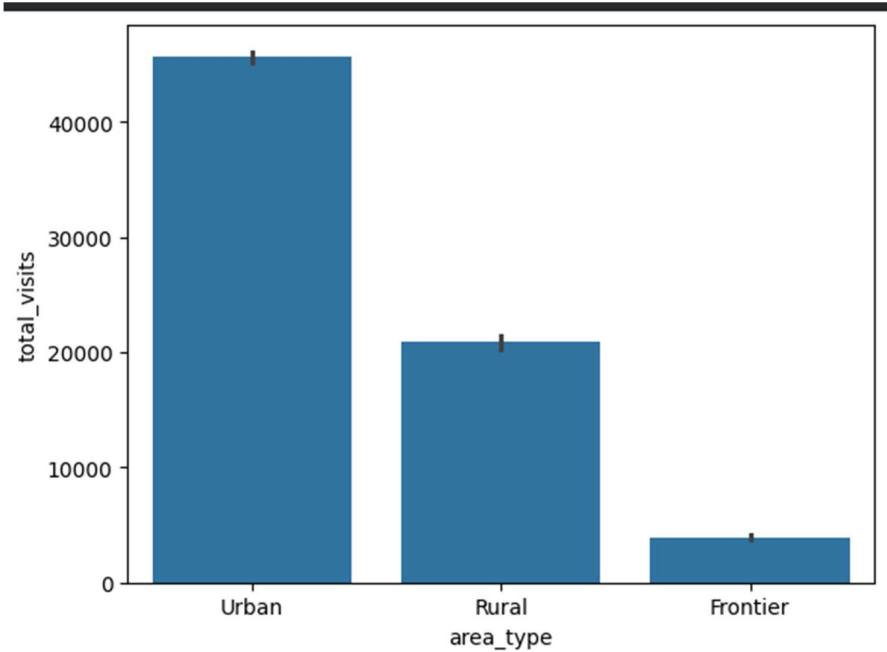
Interpretation: Condition visits < total visits, positive trend

2. Bed Size vs Total Visits



Interpretation: Larger bed size → more visits

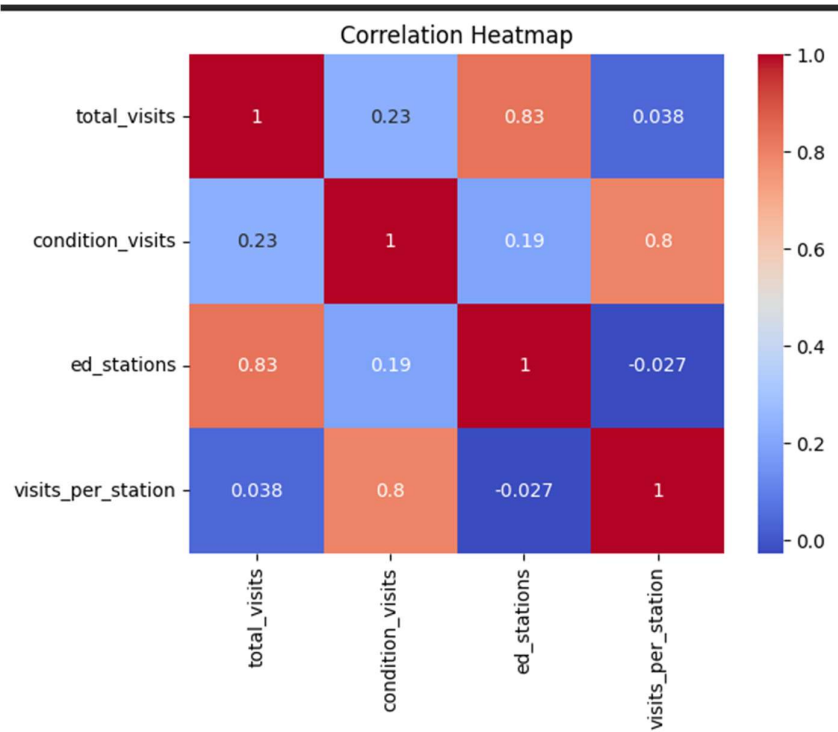
3. Area Type vs Total Visits



Interpretation: Urban has highest visits, Rural is medium, Frontier is lowest

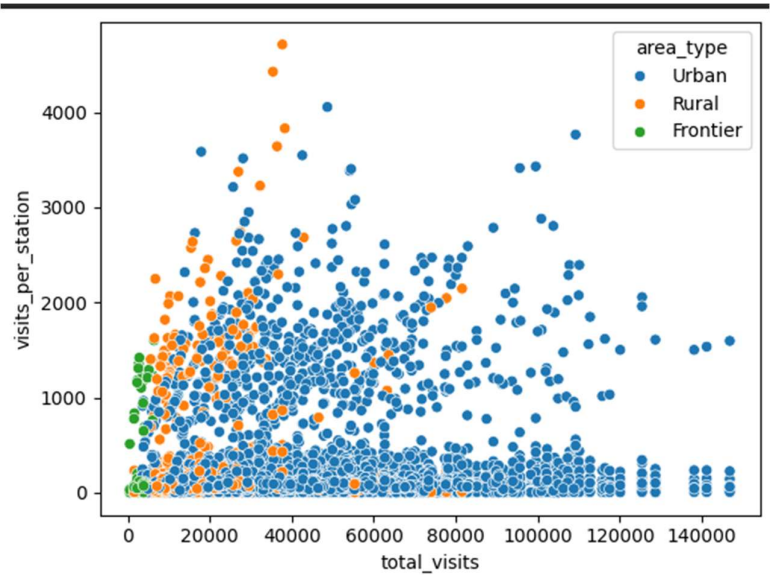
Multivariate Analysis

1. Correlation Heatmap



Interpretation: Strong relation: total_visits & ed_stations, condition_visits & visits_per_station

2. Scatter (with hue)



Interpretation: Urban → high visits spread (more usage)

Rural → medium range

Frontier → low visits mostly

Weak positive trend (visits → station usage slightly)

Types of Analysis

1.Descriptive Analysis

Total patient visits in hospitals were analyzed. It showed that 50,000 patients visited in a month, with an average of 1,600 patients per day.

2. Diagnostic Analysis

Further analysis showed that emergency visits increased mainly in urban areas due to seasonal diseases and lack of nearby clinics.

3. Predictive Analysis

Based on past trends, patient visits are expected to increase by 15% in the next month, especially in high-demand regions.

4. Prescriptive Analysis

To handle the expected increase, hospitals should add more beds, increase staff, and improve emergency department capacity.

Key Findings

- 🚦 Bigger hospitals handle more patients.
- 🚦 Urban hospitals have higher visits compared to rural & frontier.
- 🚦 Emergency cases form a major portion of total visits.
- 🚦 Some hospital systems perform significantly better than others.
- 🚦 Key variables (beds, visits, emergency) are strongly related.
- 🚦 Few hospitals handle very high volume, most are moderate/low.
- 🚦 Certain counties show higher healthcare demand.

Future Enhancements

- 📊 Forecast future patient visits using ML
- 📊 Monitor live hospital data
- 📊 Identify high-demand areas using maps
- 📊 Group patients by age, disease, region
- 📊 Improve bed & staff allocation
- 📊 Detect unusual spikes in visits
- 📊 Combine multiple hospital data sources

Conclusion

This project highlights the gap between patient demand and hospital capacity, identifying overcrowding and resource shortages. The insights help improve resource allocation, efficiency, and patient care through data-driven decisions.